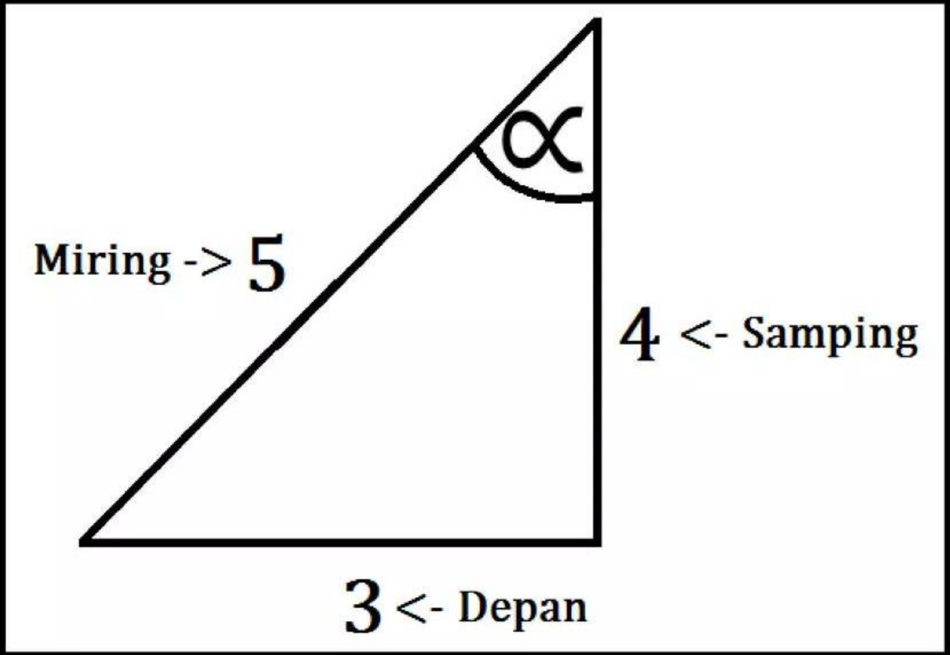
MATERI PELAJARAN PERBANDINGAN TRIGONOMETRI SMA KELAS X

Konsep Umum : PHYTAGORAS



$$\text{Tan} \propto = \frac{depan}{samping} = \frac{3}{4}$$

$$\text{Sec} \propto = \frac{1}{Cos \propto} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$$

$$\text{Cosec} \propto = \frac{1}{Sin \propto} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$$

$$\text{Sin} \propto = \frac{depan}{miring} = \frac{3}{5}$$

$$\text{Cot} \propto = \frac{1}{Tan \propto} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$$

$$\text{Cos} \propto = \frac{samping}{miring} = \frac{4}{5}$$

Perbandingan Trigonometri	Sudut-Sudut Istimewa				
	0°	30°	45°	60°	90°
Sin α	0	$^1/_2$	$^1/_2\sqrt{2}$	$^1/_2\sqrt{3}$	1
Cos α	1	$^1/_2\sqrt{3}$	$^1/_2\sqrt{2}$	$^1/_2\sqrt{2}$	0
Tann α	0	$^1/_2\sqrt{3}$	1	$\sqrt{3}$	0°

NILAI PERBANDINGAN TRIGONOMETRI

RUMUS 1
PENJUMLAHAN SUDUT
(sinus/cosinus)

RUMUS 2
JUMLAH & SELISIH SUDUT
(sinus/cosinus)

RUMUS IDENTITAS TRIGONOMETRI

RUMUS 3
PERKALIAN SUDUT
(sinus/cosinus)

RUMUS 4
TRIGONOMETRI

NILAI PERBANDINGAN TRIGONOMETRI

Kuadran II **$(180° - \propto)$** **$90° < x < 180°$** **Sin +**	**Kuadran I** **$(90° - \propto)$** **$< 90°$** **All +**
Kuadran III **$(180° + \propto)$** **$180° < x < 360°$** **Tan +**	**Kuadran IV** **$(360° - \propto)$** **Cos +**

Contoh Soal:

A. $\text{Sin } 140° = \ldots$

B. $\text{Cos } 240° = \ldots$

Penyelesaian:

A. Sin 135° = Kuadran II

$\text{Sin } (180° - \propto) = 135°$

$\text{Sin } (180° - 45°) = 135°$

$\propto = 40°$

$\text{Sin } 45° = {}^1/_2 \sqrt{2}$

Penyelesaian:

B. Cos 240° = Kuadran III

$\text{Cos } (180° + \propto) = 240°$

$\text{Cos } (180° + 60°) = 240°$

$\propto = 60°$

$\text{Cos } 60° = {}^1/_2$

RUMUS IDENTITAS TRIGONOMETRI

1. $Sin^2 A + Cos^2 A = 1$

2. $1 + Tan^2 A = \frac{1}{Cos^2 A} = Sec^2 A$

3. $1 + Cot^2 A = \frac{1}{Sec^2 A} = Cosec^2 A$

RUMUS 1
PENJUMLAHAN SUDUT (sinus/cosinus)

SINUS

1. $\text{Sin}\,(A+B) = \text{Sin}A.\,\text{Cos}B + \text{Cos}A.\,\text{Sin}B$
2. $\text{Sin}\,(A-B) = \text{Sin}A.\,\text{Cos}B - \text{Cos}A.\,\text{Sin}B$

COSINUS

1. $\text{Cos}\,(A+B) = \text{Cos}A.\,\text{Cos}B - \text{Sin}A.\,\text{Sin}B$
2. $\text{Cos}\,(A-B) = \text{Cos}A.\,\text{Cos}B + \text{Sin}A.\,\text{Sin}B$

Contoh Soal:

Tentukan Nilai dari:

A. Sin $(45° + 60°)$!

B. Sin $36°$. Cos $24°$ + Cos $36°$. Sin $24°$!

Penyelesaian:

A. $\sin(45° + 60°)$

$$= \sin 45°.\cos 60° + \cos 45°.\sin 60°$$
$$= {}^1/_2\sqrt{2}.\,{}^1/_2 + {}^1/_2\sqrt{2}.{}^1/_2\sqrt{3}$$
$$= {}^1/_4\sqrt{2} + {}^1/_4\sqrt{6}$$

B. $\sin 36°.\cos 24° + \cos 36°.\sin 24°$

$$= \sin(36° + 24°)$$
$$= \sin 60°$$
$$= {}^1/_2\sqrt{3}$$

RUMUS 2
JUMLAH & SELISIH SUDUT (sinus/cosinus)

SINUS

1. $\mathrm{Sin}A + \mathrm{Sin}B = 2.\,\mathrm{Sin}\,{}^{1}/_{2}\,(A + B).\,\mathrm{Cos}\,{}^{1}/_{2}\,(A - B)$
2. $\mathrm{Sin}A - \mathrm{Sin}B = -2.\,\mathrm{Sin}\,{}^{1}/_{2}\,(A + B).\,\mathrm{Cos}\,{}^{1}/_{2}\,(A - B)$

COSINUS

1. $\mathrm{Cos}A + \mathrm{Cos}B = 2.\,\mathrm{Cos}\,{}^{1}/_{2}\,(A + B).\,\mathrm{Cos}\,{}^{1}/_{2}\,(A - B)$
2. $\mathrm{Cos}A - \mathrm{Cos}B = -2.\,\mathrm{Sin}\,{}^{1}/_{2}\,(A + B).\,\mathrm{Sin}\,{}^{1}/_{2}\,(A - B)$

Contoh Soal:

Tentukan Nilai dari :

Cos 90° + Cos 30° !

Penyelesaian:

$$\text{Cos } 90° + Cos\ 30°$$

$$= 2. Cos\ {}^1/_2\,(A + B). Cos\ {}^1/_2\,(A - B)$$

$$= 2. Cos\ {}^1/_2\,(90° + 30°\ ). Cos\ {}^1/_2\,(90° - 30°)$$

$$= 2. Cos({}^1/_2\,.\,120°). Cos({}^1/_2\,.\,60°)$$

$$= 2. Cos(60°). Cos(30°)$$

$$= 2.\,{}^1/_2\,.\,{}^1/_2\sqrt{3}$$

$$= 1\,.\,{}^1/_2\sqrt{3}$$

RUMUS 3
PERKALIAN SUDUT (sinus/cosinus)

1. $2.\text{SinA}.\text{CosB} = \text{Sin}(A + B) + \text{Sin}(A - B)$
2. $2.\text{CosA}.\text{SinB} = \text{Sin}(A + B) - \text{Sin}(A - B)$
3. $2.\text{CosA}.\text{CosB} = \text{Cos}(A + B) + \text{Cos}(A - B)$
4. $-2.\text{CosA}.\text{CosB} = \text{Cos}\ (A + B) - \text{Sin}(A - B)$

Contoh Soal:

Tentukan Nilai dari :

$2. \cos 60°. \sin 30°$

Penyelesaian:

$2. \cos 60°. \sin 60°$

$= 2. \sin(60° + 30°) - \sin(60° - 30°)$

$= 2. \sin(90°) - \sin(30°)$

$= 2.\ 1 - {}^{1}/_{2}$

$= 2 - {}^{1}/_{2}$

RUMUS 4 TRIGONOMETRI

1. $Sin^2 \propto = 2.Sin \propto .Cos \propto$
2. $Cos^2 \propto = Cos^2 \propto - Sin^2 \propto$

   $= 2.Cos^2 \propto - 1$
3. $Tan^2 \propto = \frac{2.Tan \propto}{1 - Tan^2 \propto}$

Contoh Soal

Bentuk sederhana dari :

$Sin^2 30°$!

Penyelesaian:

$$
\begin{aligned}
Sin^2 30° &= (Sin\ 30°)^2 \\
&= (^1/_2)^2 \\
&= {}^1/_4
\end{aligned}
$$

LATIHAN SOAL

1. Cos (270°)
2. Sin (204°)
3. Tan (60°)
4. Sec (167°)
5. $\text{Sin}^2 60° + \text{Cos}^2 60°$
6. $2.\,Cos^2 60° + 2.\,Sin^2 45 + 3.\,Tan^2 45°$
7. Sin 135°
8. Cos(90°-30°)
9. Cos 60°-Cos 30°
10. Jika f(x) = (Cos x + Sin x) . Tentukan nilai f(x), jika x = 45°

PEMBAHASAN SOAL

1. Cos (270°) = K III

$$\text{Cos } (180° + \propto) = 270°$$

$$\text{Cos } (180° + \mathbf{90°}) = 270°$$

$$\propto = \mathbf{90°}$$

$$\text{Cos } \mathbf{90°} = \mathbf{0}$$

2. Sin (204°) = K III

$$\text{Sin } (180° + \propto) = 204°$$

$$\text{Sin } (180° + \mathbf{24°}) = 204°$$

$$\propto = \mathbf{24°}\ (-)$$

3. Tan (60°) = K I

$$\text{Tan } (90° - \propto) = 60°$$

$$\text{Tan } (90° - \mathbf{30°}) = 60°$$

$$\propto = \mathbf{30°}$$

$$\text{Tan } 30° = {}^{1}/_{3}\sqrt{3}$$

4. Sec (167°) = K II

$$\text{Sec } (180° - \propto) = 167°$$

$$\text{Sin } (180° - \mathbf{13°}) = 167°$$

$$\propto = \mathbf{13°}\ (+)$$

5. $\text{Sin}^2 60° + \text{Cos}^2 60°$

$= (\text{Sin } 60°)^2 + (\text{Cos } 60°)^2$

$= (\frac{1}{2}\sqrt{3})^2 + (\frac{1}{2})^2$

$= \frac{1}{4} . 3 + \frac{1}{4}$

$= \frac{3}{4} + \frac{1}{4}$

$= \frac{4}{4}$

$= 1$

6. $2.\ \text{Cos}^2 60° + 2.\ \text{Sin}^2 45 + 3.\ \text{Tan}^2 45°$

$= 2(\text{Cos } 60°)^2 + 2(\text{Sin } 60°)^2 + 3(\text{Tan } 45°)^2$

$= 2(\frac{1}{2})^2 + 2(\frac{1}{2}\sqrt{2})^2 + 3(1)^2$

$= 2(\frac{1}{4}) + 2(\frac{1}{4}.2) + 3.1$

$= \frac{2}{4} + \frac{2}{4}.4 + 3$

$= \frac{2}{4} + 2 + 3$

$= \frac{2}{4} + 5$

7. $\text{Sin } 135° = (\text{Sin } 90+45)$

$= \text{Sin}\,(A + B) = \text{Sin}A.\text{Cos}B + \text{Cos}A.\text{Sin}B$

$= \text{Sin } 90°.\text{Cos } 45°+\text{Cos } 90°.\text{Sin } 45°$

$= 1.{}^{1}/_{2}\sqrt{2}+0.{}^{1}/_{2}\sqrt{2}$

$= {}^{1}/_{2}\sqrt{2}$

8. $\text{Cos}(90°\text{-}30°)$

$= \text{Cos}\,(A - B) = \text{Cos}A.\text{Cos}B + \text{Sin}A.\text{Sin}B$

$= \text{Cos } 90°.\text{Cos } 45°+\text{Sin } 90°.\text{Sin } 45°$

$= {}^{1}/_{2}\sqrt{2}.0+{}^{1}/_{2}\sqrt{3}.1$

$= {}^{1}/_{2}\sqrt{3}$

9. Cos 60°-Cos 30°

$= \mathbf{CosA - CosB = -2. Sin\ ^1/_2\,(A + B). Sin\ ^1/_2\,(A - B)}$

$= \mathbf{-2. Sin\ ^1/_2\,(60° + 30°). Sin\ ^1/_2\,(60° - 30°)}$

$= \mathbf{-2. Sin\ ^1/_2\,(90°). Sin\ ^1/_2\,(30°)}$

$= \mathbf{-2. Sin\ 45°. Sin\ 30°}$

$= \mathbf{-2.\ ^1/_2\sqrt{2}. Sin\ 15°}$

$= \mathbf{-\ ^2/_2\,. -2. Sin\ 15°}$

$= \mathbf{-1. -2. Sin\ 15°}$

$= \mathbf{2. Sin\ 15°}$

10. Jika f(x) = (Cos x + Sin x) . Tentukan nilai f(x), jika x = 45°

F (x) = (Cos x + Sin x) (Cos x + Sin x)

$= Cos^2 x + Sin\, x.\,\mathrm{Cos}\ x + Sin\, x.\,\mathrm{Cos}\ x + Sin^2 x$

$= Cos^2 x + Sin^2 x + Sin\, x.\,\mathrm{Cos}\ x + Sin\, x.\,\mathrm{Cos}\ x$

$= 1 + Sin\, x.\,\mathrm{Cos}\ x + Sin\, x.\,\mathrm{Cos}\ x$

= 1 + (Sin (A+B)+Sin (A-B))

= 1 + (Sin (45°+45°)+Sin (45°-45°))

= 1 + (Sin 90° +Sin90°)

= 1 + 1 + 0

= 2

SELAMAT BELAJAR